

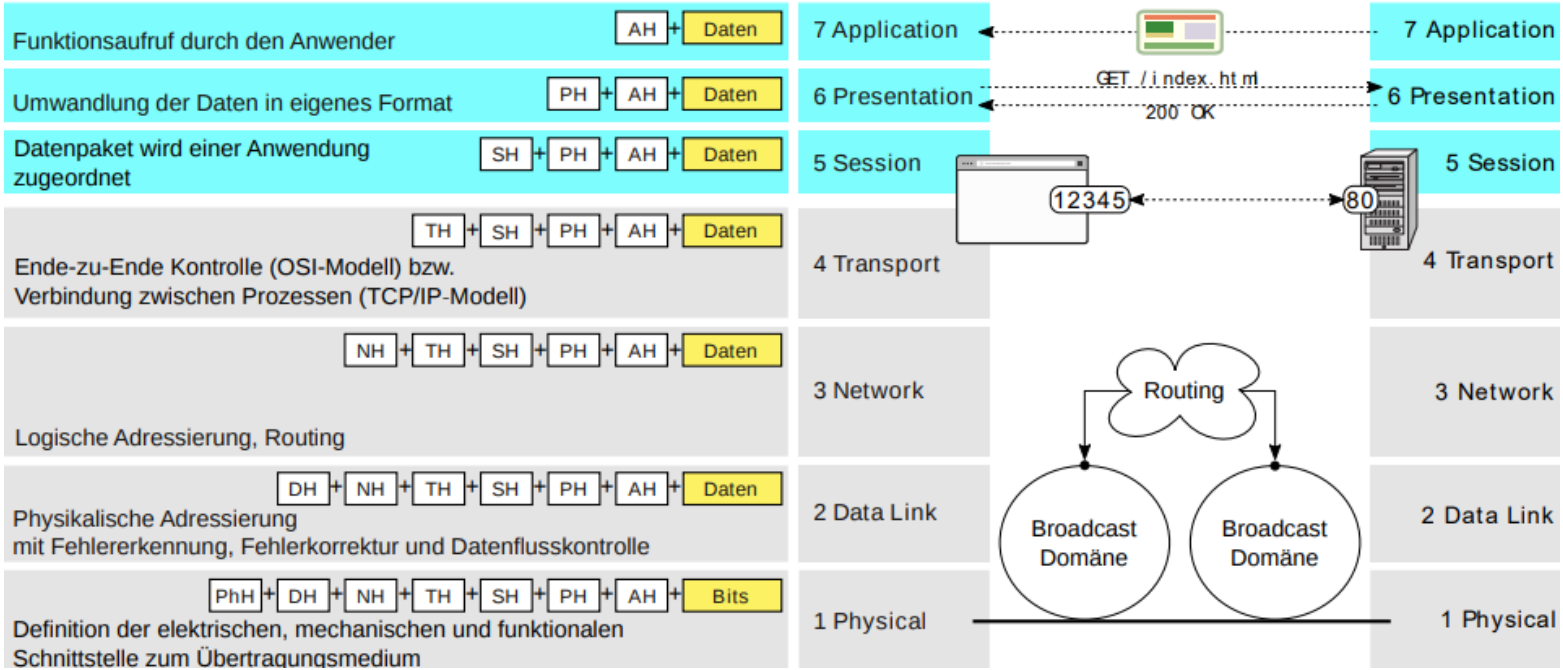
IT Security Basics

Secure Communications

Classification "Secure communication"

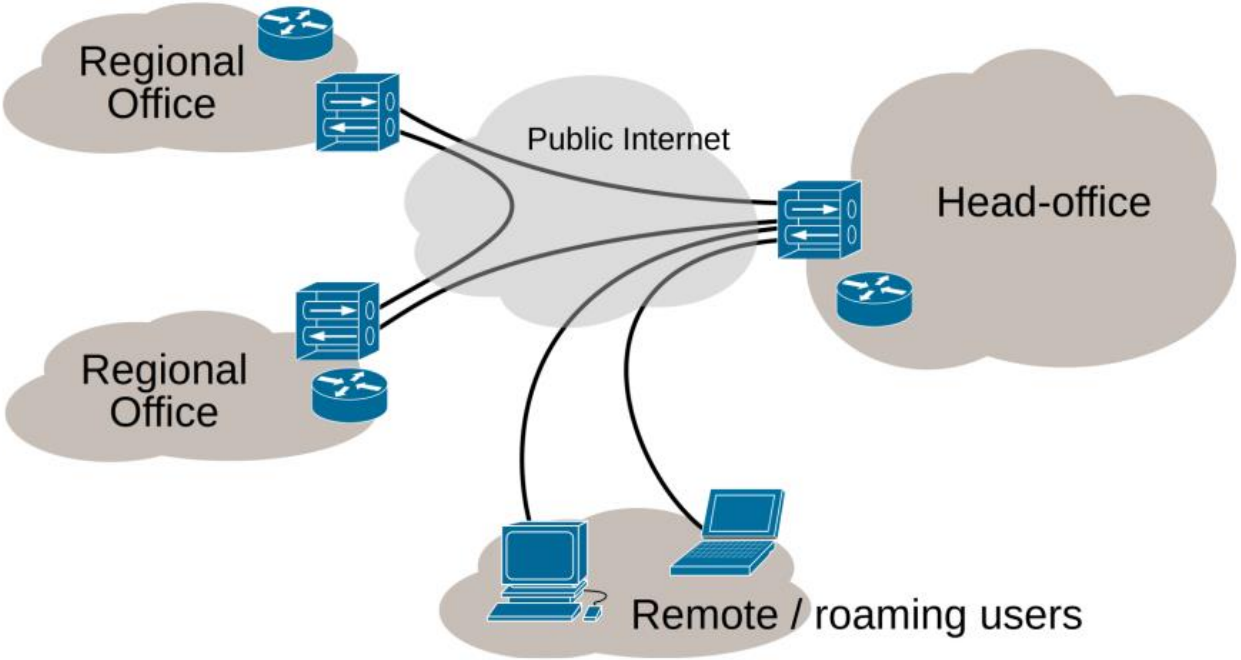
- **Secure communication protocols**
  - Safe channel between source and destination
  - Introductory connection setup for parameter agreement and (if necessary, mutual auth)
  - IPsec, TLS, SSH, ...
- **Integrated security mechanisms**
  - Explicit application or integrated in application
  - one-pass processing
  - PGP, S/MIME, ...

OSI model



Johannes Stoll and A.Ph.Lintenhofer. https://commons.wikimedia.org/wiki/File: OSI7Layer_model_3.1.svg), OSI7Layer model 3.1, Revised and extended by A.Ph.Lintenhofer, https://creativecommons.org/licenses/by-sa/4.0/legalcode, Nov 2019

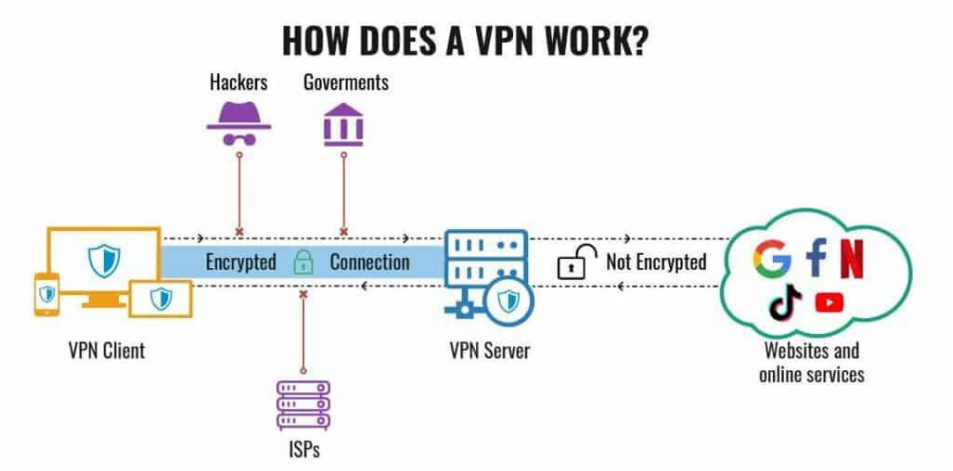
VPN - Virtual Private Network



Ludovic.ferre. (talk contribs) (https://commons.wikimedia.org/wiki/File:Virtual_Private_Network_overview.svg), Lettering and layout by A.Ph.Lintenhofer, https://creativecommons.org/licenses/bysa/4.0/legalcode, Nov 2019

VPN

- **Virtual Private Network**
- **Establishes secure communication**
  - Creates point-to-point tunnel that encrypts and protects data
  - Allows secure communication in a insecure network (e.g. Internet)
  - No physical cables needed



Source: https://www.comparitech.com/blog/vpn-privacy/vpn-tunneling/

VPN

- **Reasons for VPN**
  - Secure data while e.g. work from home
  - Access or stream regional content from anywhere
  - Bypass censorship and surveillance
- **VPN solutions (examples)**
  - WireGuard – fast and modern, simple to configure, good when using mobile devices
  - OpenVPN – highly secure, open source, good for remote working
  - IPSec – highly secure, complex to configure, used for site-to-site VPN
    - https://www.ivpn.net/en/pptp-vs-ipsec-ikev2-vs-openvpn-vs-wireguard/
    - https://www.avast.com/c-vpn-protocols



Source: https://1000logos.net/best-vpn-services/

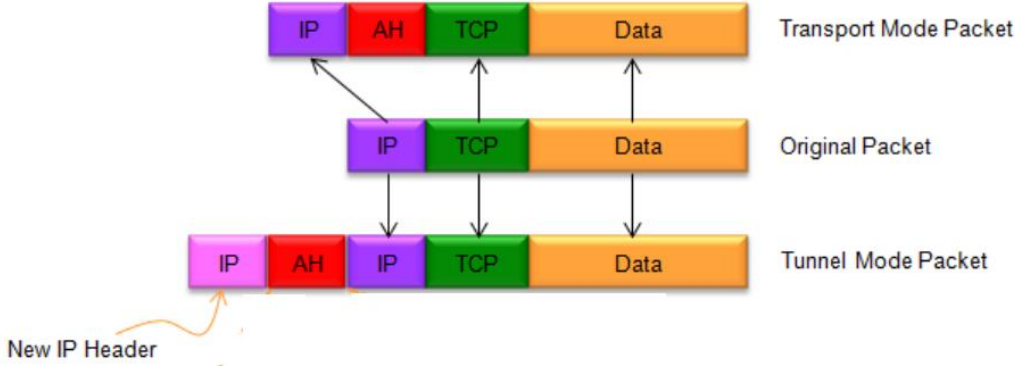
VPN - IPSec

- **One way to create a VPN**
- **Properties:**
  - Transparent to the application itself
  - Works on the network layer of the OSI Modell
  - Connects hosts and/or networks

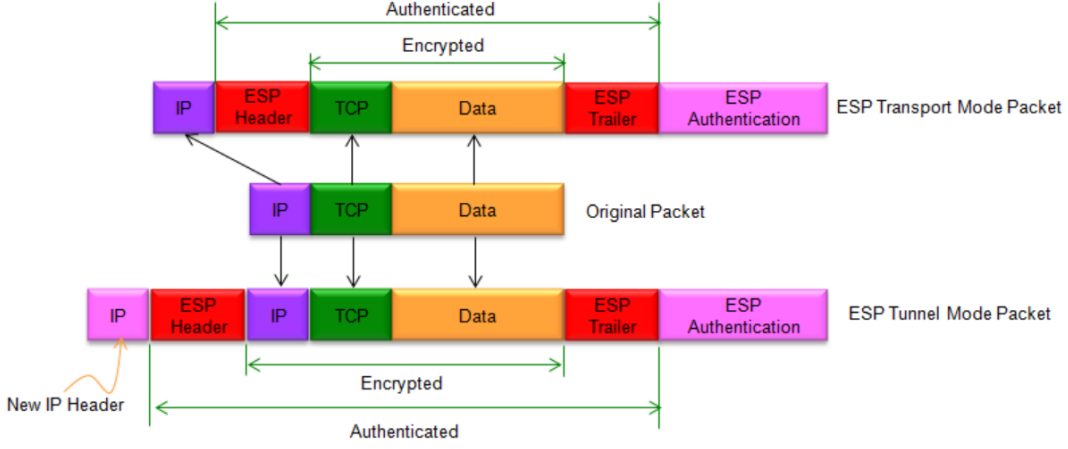
- **Two different modes**
  - Authentication Header
  - Encapsulating Security Payload

VPN - IPSec

- **Authentication Header (AH)**
  - Protects:
    - Authenticity
    - Integrity
    - Against Replay Attacks

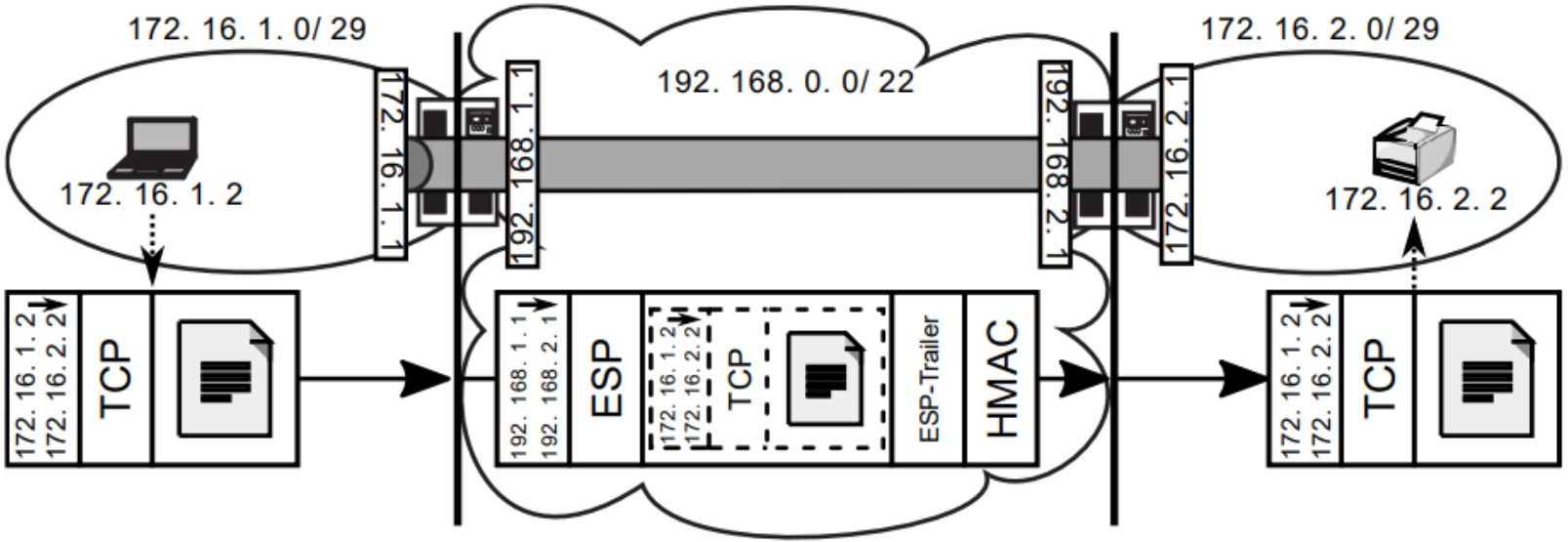


- **Encapsulating Security Payload (ESP)**
  - Protects:
    - Authenticity
    - Integrity
    - Confidentially
    - Against Replay Attacks



Sources: https://www.sharetechnote.com/html/IP_Network_IPSec_AH.html
https://www.sharetechnote.com/html/IP_Network_IPSec_ESP.html

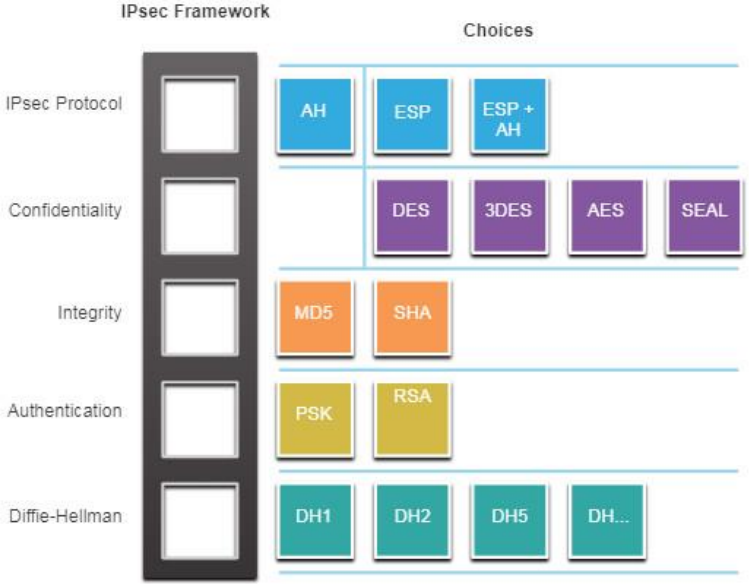
IPsec in tunnel mode
Protocol: ESP

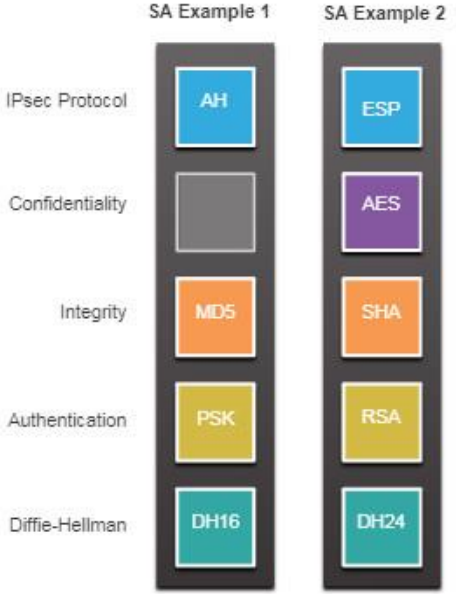


- Tunnel mode: Protects the whole package (not only the message)

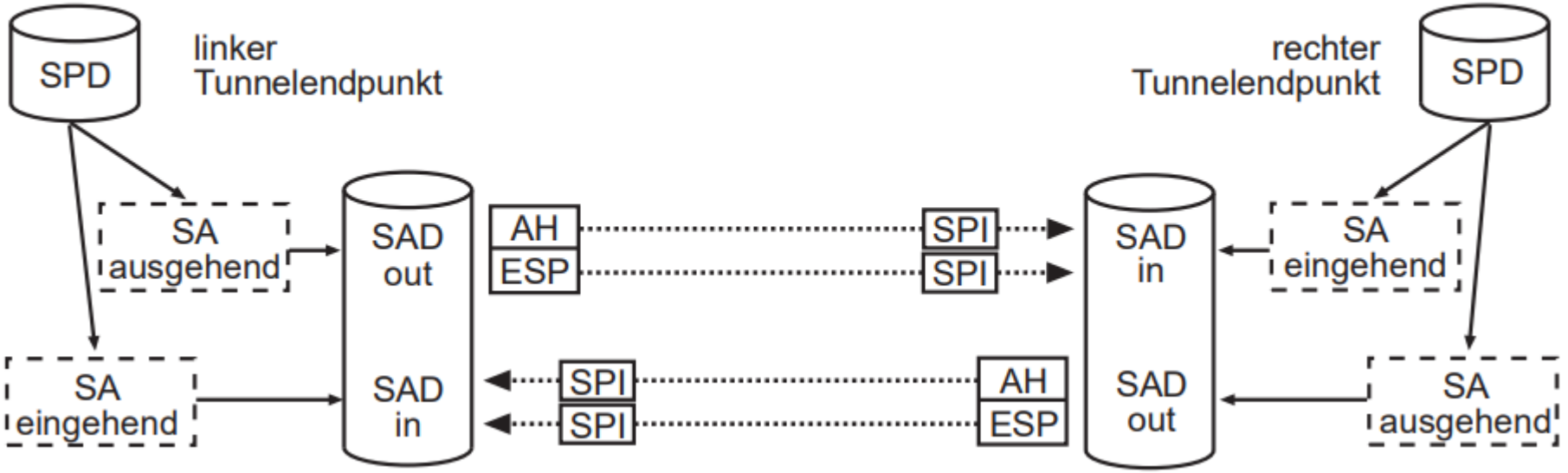
VPN - IPSec

- **Used algorithms**
  - Many algorithms supported
  - Which to use: Negotiated when creating the tunnel

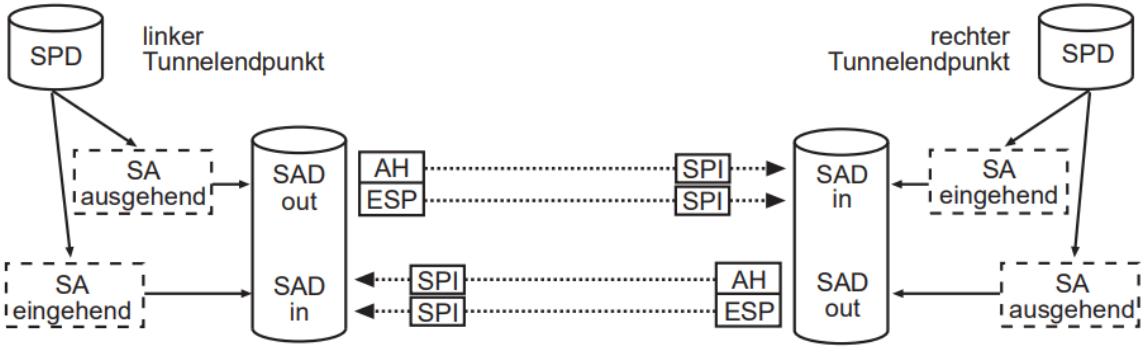




IPsec - Administration of the security parameters



IPsec - Administration of the security parameters

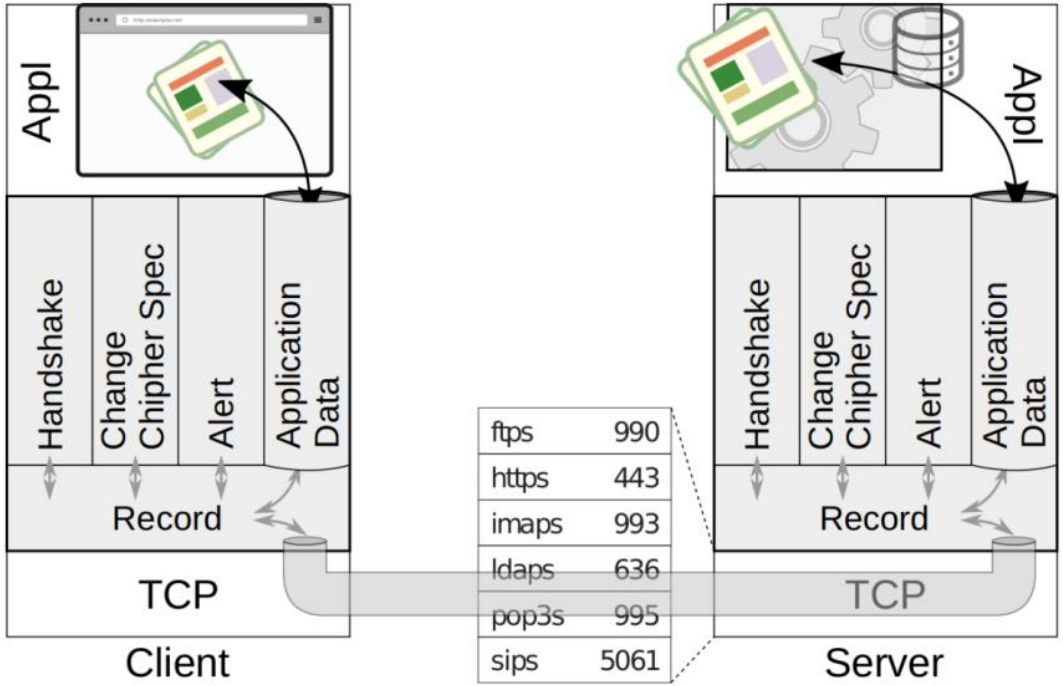


- SP: Security Policy - connection-specific, which protocols are used
- SPD: Security Policy Database - stores all SP for all connections, determines which protection level the traffic has
- SA: Security Association – instance of a SP, a specific connection within an network with all its parameters
- SAD: Security Association Database – database to store all SA
- SPI: Security Parameter Index – index of the SA that is stored in the SAD, within all packages

TLS – Transport Layer Security

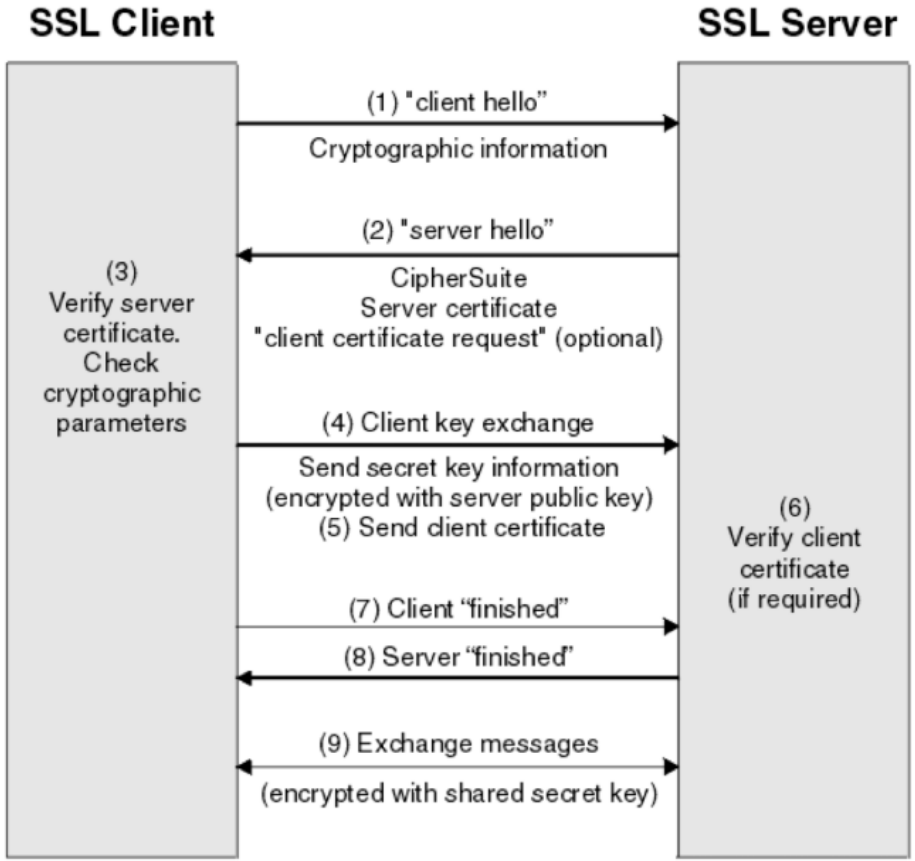
- **Encrypts traffic between client and server**
- **Properties:**
  - Independent from applications
  - Works on presentation and session layer of the OSI Model
  - Needs to agree parameters as part of the protocol
  - Connects processes
- **Protects CIA**
- **Two parts:**
  - TLS Handshake Layer – setting parameters of the connection
  - TLS Record Layer – exchanging data

TLS - Transport Layer Security



- TLS Handshake Layer:
  - Handshake Protocol: Session setup, setting parameters
  - Change Cipher Spec: Change of parameters
  - Application Data: Sends data to TLS Record Layer
  - Alert: Alerts for connections

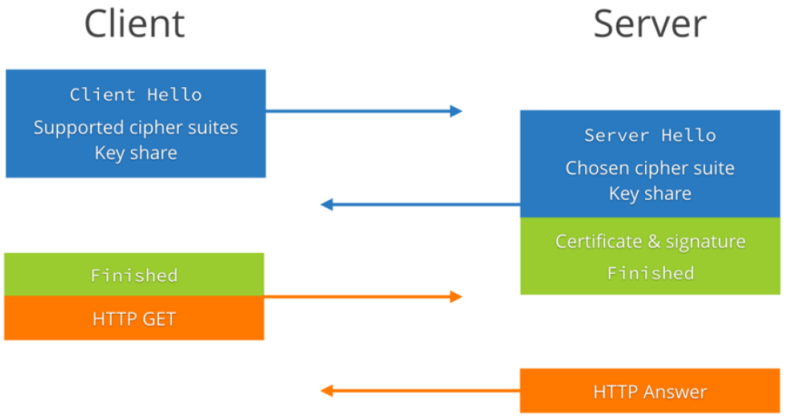
TLS - Handshake 1.2

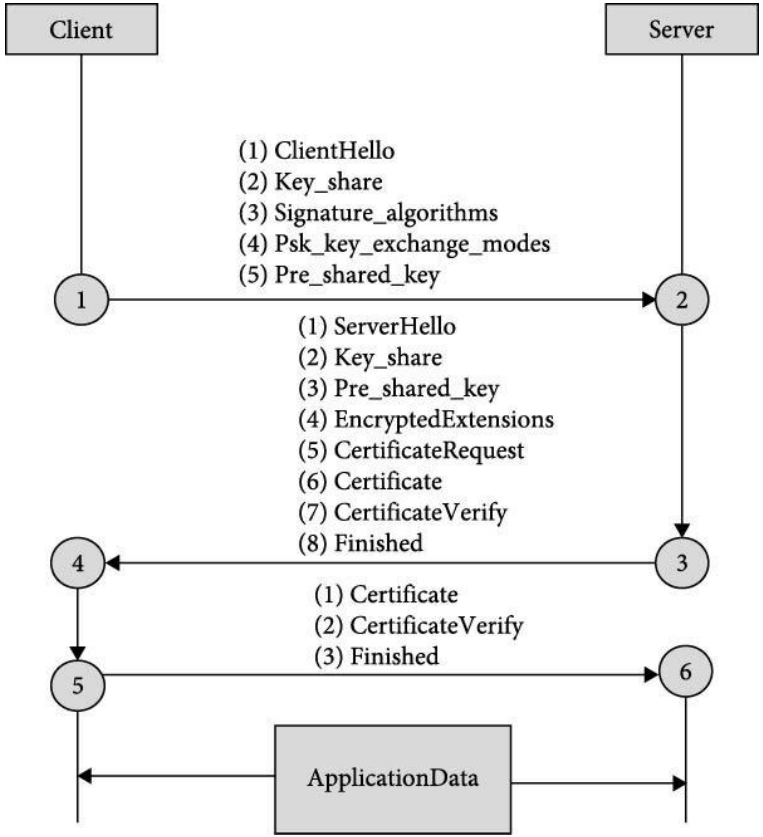


Source: https://venafi.com/blog/where-tlsssl-handshake-most-vulnerable/

TLS - Handshake 1.3

- **Properties:**
  - Shorter, only 2 steps necessary
  - Already sends data with the finish message
  - Sends key information directly with the hello

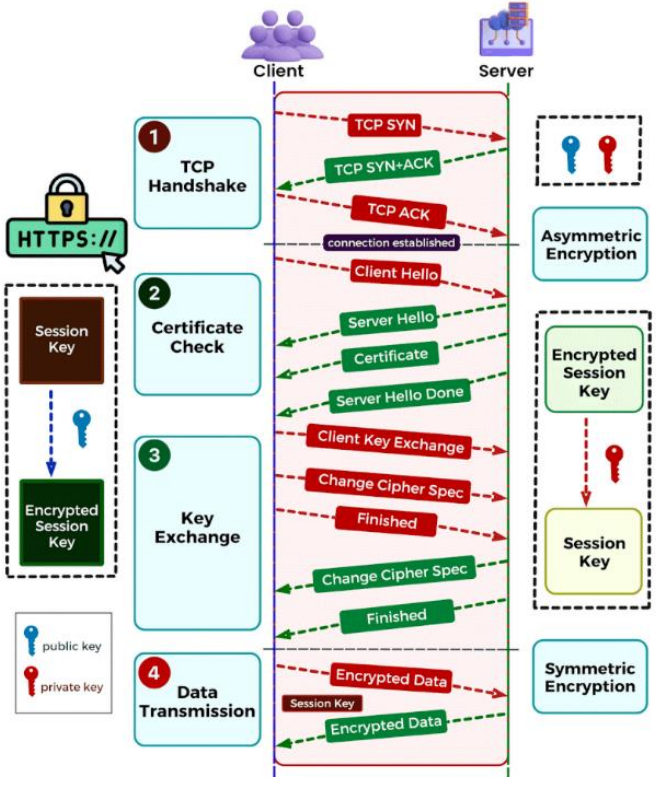


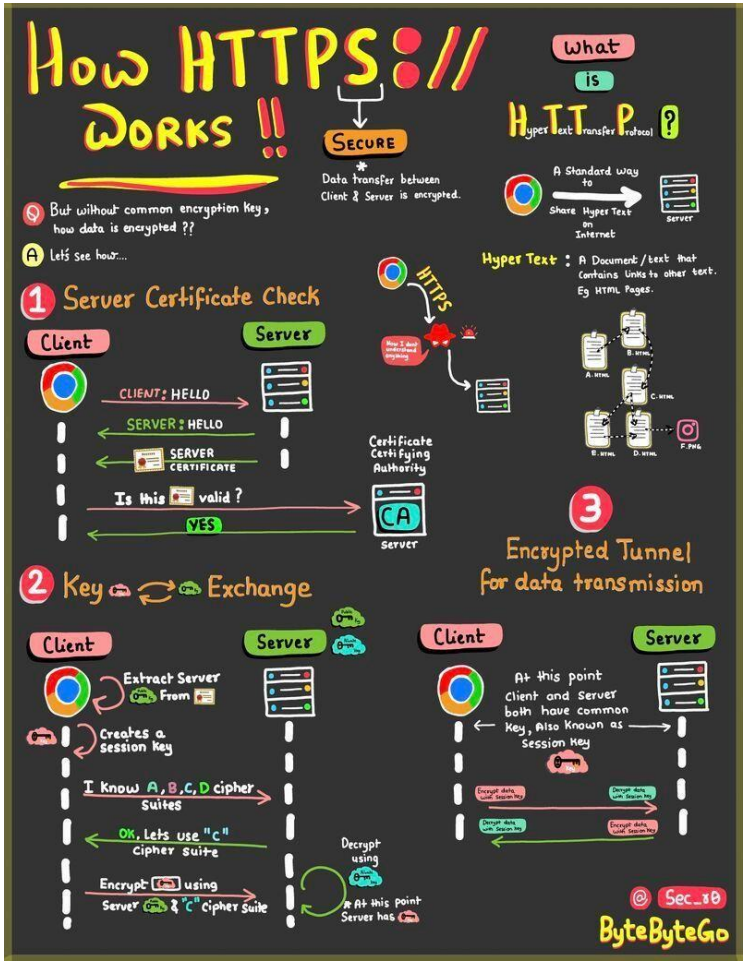


Source: https://venafi.com/blog/where-tlsssl-handshake-most-vulnerable/
https://www.researchgate.net/figure/Node-labeling-during-the-TLS13-handshake-interaction_fig5_381396369

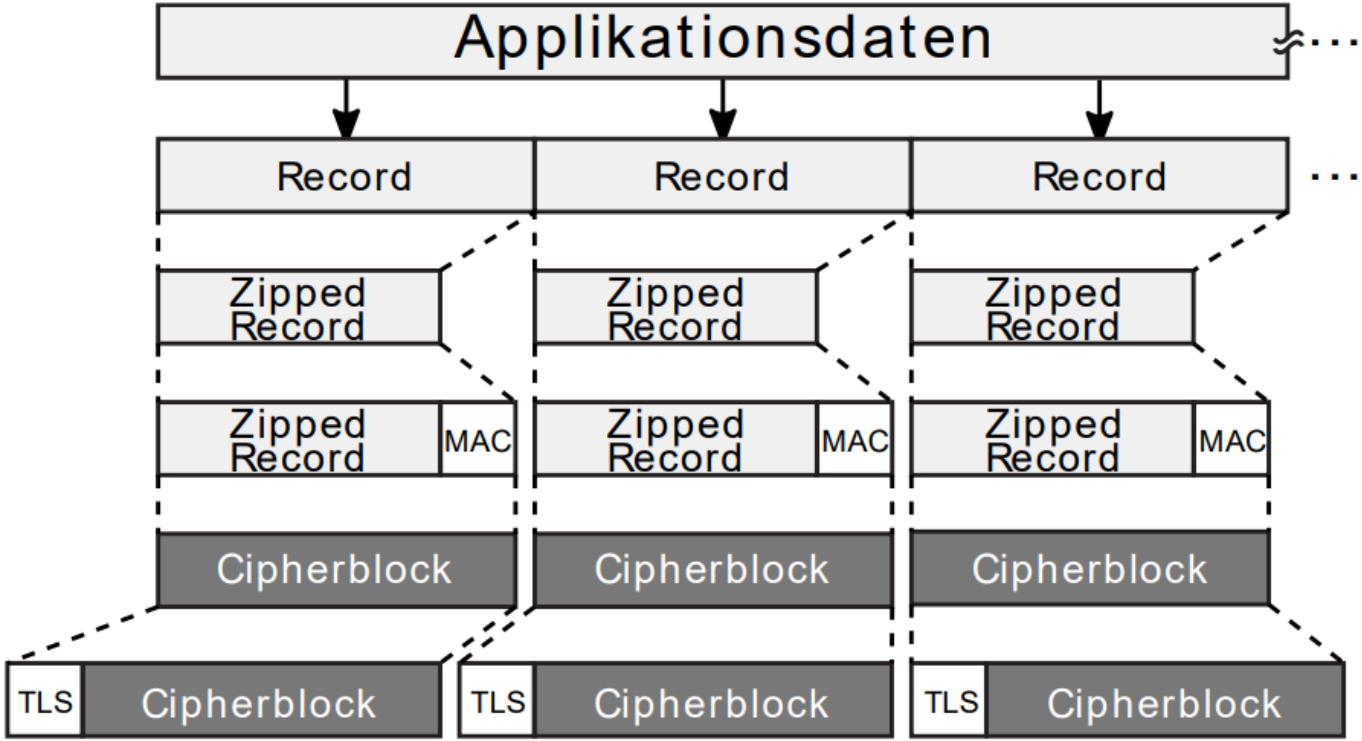
TLS - Example

- **Example for TLS:**
  - HTTPS

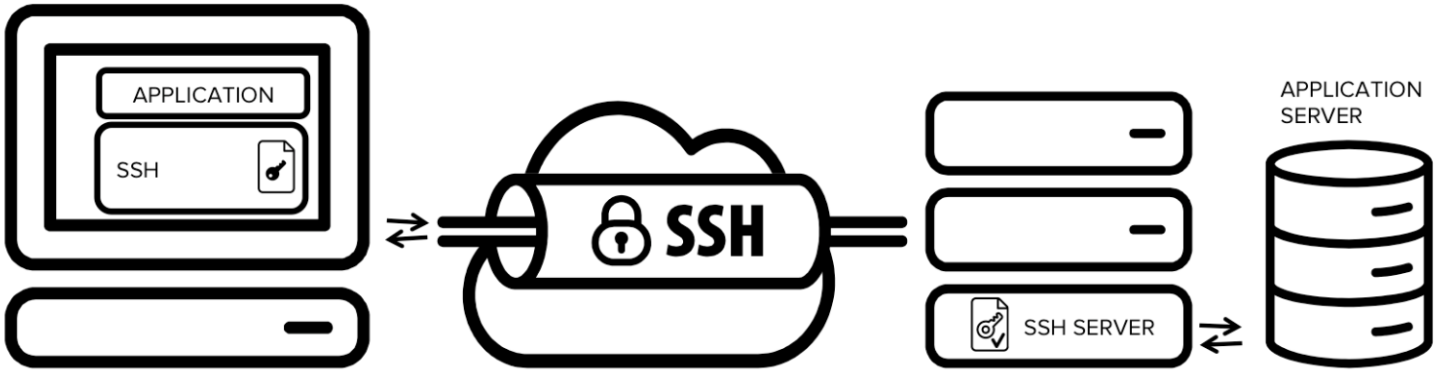




TLS - Record Layer



SSH - Secure Shell



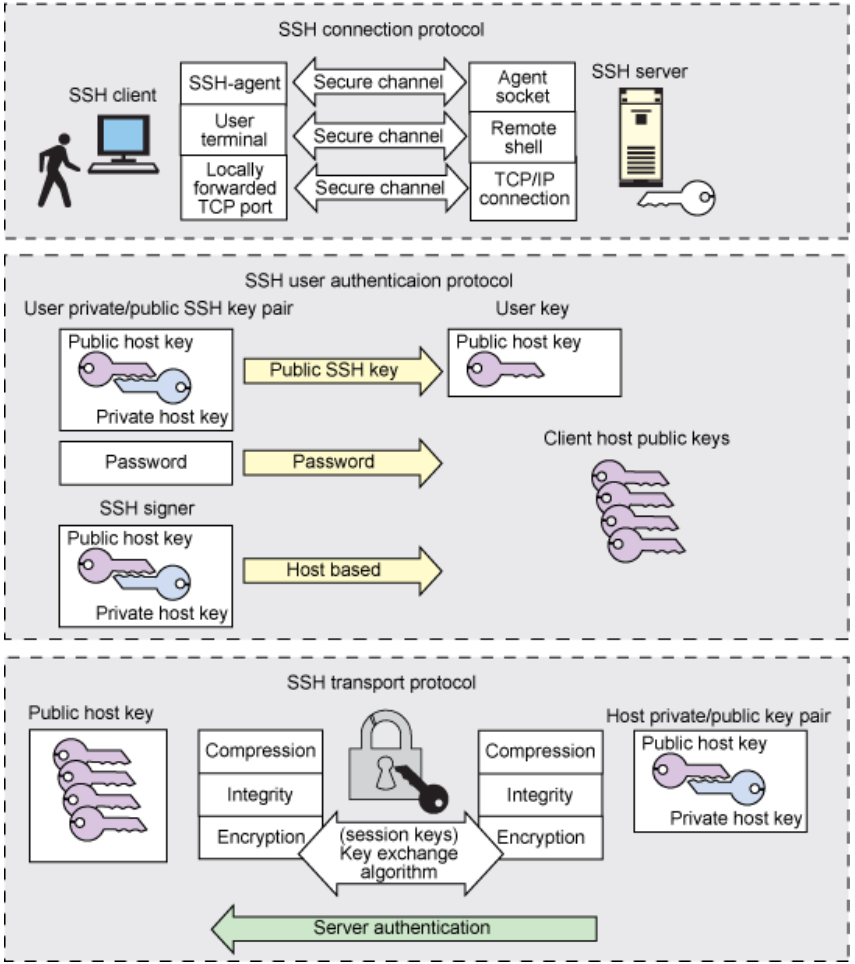
Source: https://www.ssh.com/academy/ssh/tunneling

SSH - Secure Shell

- **Properties:**
  - Based on TCP
  - Allows remote login and execution of commands
  - Sends data encrypted over a tunnel
- **Protects CIA**
- **Three Layers:**
  - Transport Layer
  - User Authentication Layer
  - Connection Layer

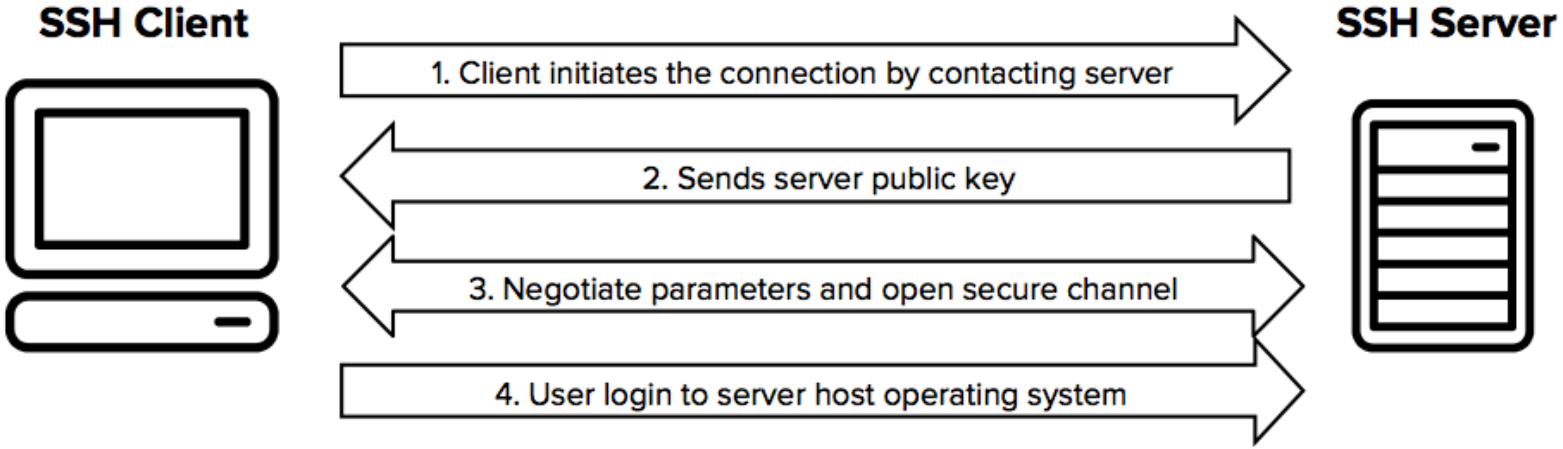
SSH - Secure Shell

- **Transport Layer**
  - Lowest Layer
  - Negotiation of parameters
  - Creates a cryptographic connection
- **User Authentication Layer**
  - Authentication of the user
  - Methods: e.g. password, public key, etc.
- **Connection Layer**
  - Many channels in one connection possible
  - Different services in different channels possible



Quelle: https://developer.ibm.com/developer/default/articles/au-sshsecurity/images/SSH_public_private_key.gif

SSH - Authentication



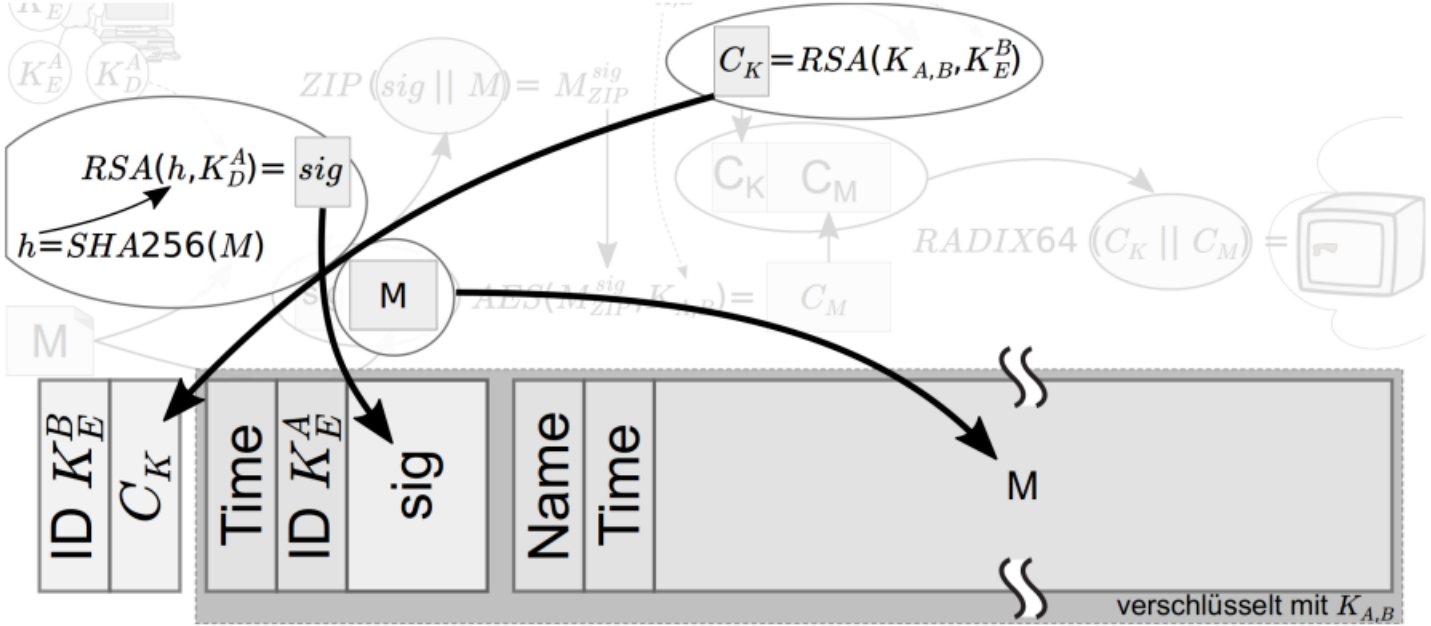
Source: https://www.ssh.com/academy/ssh

PGP

- **"Pretty Good Privacy" (PGP):**
  - Independent application
  - Encrypts files or e-mails
  - Protection goals for files:
    - Confidentiality
    - Integrity
    - Authenticity
- **Results:**
  - Encrypted/signed container
  - As a file with attached parameters
  - = One Pass Processing
    - Streaming algorithm, reads Data only once

PGP - Pretty Good Privacy

- Asymmetric encryption using public (K_E) and private key (K_D) for key exchange
- Signature to verify sender using public (K_E) and private key (K_D)
- Symmetric encryption for communication using session keys ($K_{A,B}$)



S/MIME

- **Secure/Multipurpose Internet Mail Extensions**
  - RFC-standard for e-mail encryption
  - Based on certificates and PKI
  - Integrated in most of the common e-mail programs
  - Secure version of MIME for attachments like audio, video, images or other types of media
- **Functionality**
  - Usage of digital signatures to verify the sender of a message
  - Uses asymmetric encryption to protect the content of the messages

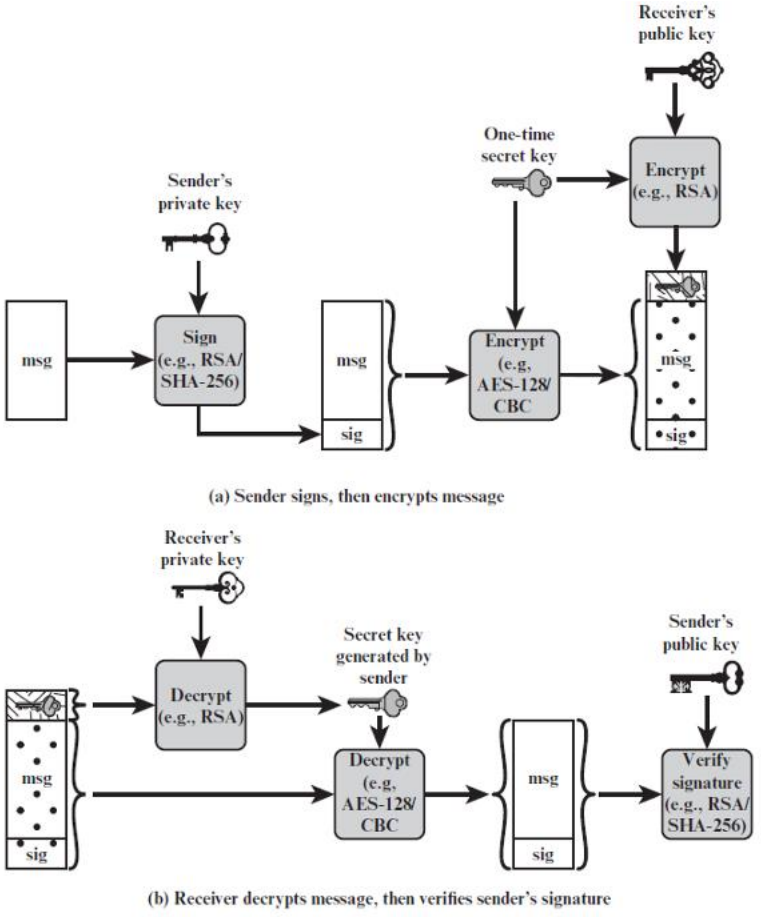


Fig. 3: S/MIME Flow [19, p. 631]

Source: https://pub.fh-campuswien.ac.at/obvfcwhsacc/content/titleinfo/7872885/full.pdf